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Test Format, Learning Confidence, and Perceptions of Teaching Effectiveness

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Abstract

Background: Both multiple-choice and short-answer tests can be beneficial to learning in the classroom. However, fact-based multiple-choice questions, because they include the correct answer as a response option, could lead to inflated estimates of learning and higher evaluations of teaching effectiveness.
Objective: The objective of this study was to examine the effects of test format on perceptions of learning and teaching.
Method: Undergraduate students ($N = 123$) completed either a multiple-choice or short-answer test based on a brief psychology lesson. Then, they predicted their performance on later tests of knowledge and evaluated the quality of the lesson.
Results: Taking a multiple-choice test led to predictions about future performance that were 10% higher than those taking a short-answer test. No consistent differences emerged in participants' evaluations of teaching effectiveness between the two test formats.
Conclusion: These results suggest that test format may significantly influence students' confidence in their ability to produce answers on future tests.
Teaching Implications: Teachers who want to take advantage of testing as a metacognitive learning tool should adopt question formats that utilize recall of information from memory without retrieval cues.

Keywords

testing, college teaching, metacognition, student evaluation of teaching

Testing students to help them learn is a standard, evidence-based practice in psychology (Dunlosky & Rawson, 2015; Michaels, 2017; Narloch et al., 2006). Having students retrieve knowledge using multiple-choice or short-answer test questions can significantly increase learning (Agarwal et al., 2021; McDermott et al., 2014). Testing directly improves memory through retrieval practice, but it can have indirect benefits as well. Answering test questions allows students to become aware of gaps in their knowledge, which is an essential form of metacognition (Rivers, 2021). Without such feedback, fluency with information and the feeling of knowing may lead students to develop false confidence in their understanding (R. A. Bjork et al., 2013). Thus, testing is one of the desirable difficulties that make initial learning seem more difficult while increasing long-term retention (E. L. Bjork & Bjork, 2011). Test format could impact this desirable difficulty. Because multiple-choice questions that assess factual knowledge require recognition rather than recall, responding is easier and more fluent than for short-answer questions. As such, fact-based multiple-choice tests may lead students to overestimate their knowledge because obtaining correct answers requires relatively shallow cognitive processing and correct answers can be discerned even when understanding is incomplete.

Successful learning requires students to self-assess their understanding of course content, but metacognition research shows that the use of outside assistance to answer questions leads to overconfidence in knowledge (Fisher et al., 2015; Fisher & Oppenheimer, 2021a, 2021b). For example, when answering general knowledge questions as part of a team, people inflate estimates of how many questions they could correctly answer when working alone (Fisher & Oppenheimer, 2021b). People also inflate estimates of knowledge when searching the internet for answers to questions rather than answering the questions from memory (Fisher et al., 2015). Overall, testing that relies on outside assistance creates an illusion of knowledge by preventing people from gaining metacognitive insight into gaps in learning.

Test questions themselves can be a form of outside assistance. For example, questions that provide hints about correct answers inflate peoples' estimates of their ability to answer questions without such cues (Fisher & Oppenheimer, 2021a). In one study of this effect with direct implications for teaching, participants memorized 15 words and then completed either a free-recall test with no outside assistance or a fill-in-the-blank test where the first three letters of each word were visible as prompts (Fisher & Oppenheimer, 2021a; Study 2a). The average number of words correctly recalled on the fill-in-the-blank test with prompts was 62%, but the average on the free-recall test was only 27%—free recall was significantly more difficult. However, when participants predicted how they would perform on a future free-recall test without hints or outside assistance,

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participants who completed the fill-in-the-blank test with prompts believed they would recall significantly more words (48%) than participants who completed the free-recall test (26%) – fill-in-the-blank questions with prompts led to significant overconfidence about free-recall performance. Similar to fill-in-the-blank questions with prompts, fact-based multiple-choice test questions provide outside assistance because they include response options. As such, multiple-choice tests could lead students to overestimate their knowledge relative to short-answer tests.

Test format may also affect evaluations of teaching effectiveness. Generally, students prefer multiple-choice tests over other test formats (Holzinger et al., 2020; Neto et al., 2022; Struyven et al., 2005). One explanation for the preference is students' belief that multiple-choice tests are easier than other types of tests (Holzinger et al., 2020; Parkes & Stefanou, 2010; Struyven et al., 2005). Study habits reflect this belief because students spend more time and effort preparing for short-answer and essay tests than multiple-choice tests (Narloch et al., 2006; Scouller, 1998; Struyven et al., 2005). Because they require more work, students may dislike the challenge associated with short-answer and essay tests. When students view instruction as too challenging, they evaluate it negatively (Addison et al., 2006; Centra, 2003). Although students do not want their courses to be overly difficult, they do want to learn. Perceptions of learning are correlated with positive teaching evaluations (Benton & Li, 2015; Stehle et al., 2012). Thus, if multiple-choice tests lead students to overestimate how much they have learned, this perception of learning, even if illusory, could increase evaluations of teaching.

The purpose of the current study was to explore the relations between test format, perceptions of learning, and evaluations of teaching. Undergraduate students in the study watched a brief psychology lecture and then completed a multiple-choice or short-answer test designed to assess knowledge from the lesson (i.e., the remembering level of Bloom's taxonomy, Krathwohl, 2002). After receiving the correct answers and self-scoring the test, students predicted how they would perform on future tests and evaluated the quality of the lecture. Based on these methods, there were five research questions.

1. Do students predict that they will perform better on multiple-choice tests than on tests that require them to write down the answers?
2. Does test format affect students' predictions about future performance?
3. Does test format affect students' perceptions of teaching effectiveness?
4. Is test performance related to students' predictions about future performance?
5. Is test performance related to students' perceptions of teaching effectiveness?

The answers to these questions will help teachers understand how choices about test format could impact students' perceptions of teaching and learning.

Method

Participants

The sample ($N = 123$) consisted of undergraduate students from a small, private university in the Midwestern U.S. The sample was mostly female (63%, $n = 77$; male = 36%, $n = 44$; nonbinary = 2%, $n = 2$) and White (78%, $n = 96$; multiethnic = 13%, $n = 16$; African American = 4%, $n = 5$; Hispanic or Latino = 2%, $n = 3$; Asian = 2%, $n = 2$; other = 1%, $n = 1$). The average age of participants was 21 ($SD = 4$), and the sample was equally distributed across years in college (first-year = 23%, $n = 28$; second-year = 24%, $n = 28$, third-year = 23%, $n = 28$; fourth- and fifth-year = 30%, $n = 37$). The modal number of psychology courses taken by participants was 1 ($M = 3$, $SD = 1$). Researchers recruited participants by visiting classes across the university during the spring term of 2023 and asking students in attendance to participate. The project received Institutional Review Board approval. The materials, data, and syntax that support the findings of this study are openly available online (Boysen & Osgood, 2023).

Materials and Procedure

This study conceptually replicated past research on the metacognitive effects of outside assistance (Fisher & Oppenheimer, 2021a). The basic procedure consisted of a brief psychology lesson, test, and survey administered to groups of students in classrooms (average group size was approximately 15). Researchers informed students that they would be participating in a research study about education and that they would watch a brief video lecture and then take a quiz on the content. The instructions asked participants to pay attention and try on the quiz just like they would in a college class. Participants watched a 3-minute lecture video consisting of PowerPoint slides with a voiceover and closed captioning. The lecture outlined the basic principles of the stereotype content model with an emphasis on the concepts of interpersonal warmth and competence (Cuddy et al., 2007; Fiske et al., 2002). This lecture topic represented a theory that students would not have encountered in the University's psychology curriculum.

After watching the video, participants completed a demographic survey consisting of five questions. Instructions on the survey indicated that participants should not go on to the next section until instructed to do so. The purpose of placing the demographic survey after the video was to clear participants' working memory of content from the lesson. Reading and answering demographic questions should prevent rehearsal of information and increase forgetting due to interference (Jonides et al., 2007). After all participants had time to complete the demographic survey (2–3 min), the researchers administered a 5-item test about facts from the lesson. The tests assessed knowledge from the lesson at the remembering level of Bloom's taxonomy (Krathwohl, 2002). Random assignment, achieved by alternating surveys while handing them out, placed participants into one of two conditions: multiple-choice test or short-answer test. Question prompts were identical between the two conditions and included items such as "Who was the creator of the stereotype content model" and "What are the key characteristics associated with perceptions of high warmth." In the multiple-choice condition, participants had to select the correct answer from four response options. In the short-answer condition, participants had to write down the correct answer. Participants could not collaborate, use notes, or look up answers.

Participants had about 5 min to complete the test, then the researchers read the text of the correct answers so that participants in both conditions could score their tests. For example, the researchers stated, "For the question 'What are the key characteristics associated with perceptions of high warmth'. The correct answer is friendliness, trustworthiness, sincerity." Participants scored their own tests and recorded their total number of correct responses. The researchers instructed participants to count only items as correct if they had the complete answer and not to assign credit for partial answers. The researchers did not check the accuracy of participants' self-scores during the procedures. However, once the study was complete, the researchers evaluated the accuracy of participants' test answers to produce a total score out of five points. Single researchers scored the tests by comparing responses to an answer key and assigning one point for exact matches (e.g., in response to "the key characteristics associated with warmth," an answer of "friendliness, trustworthiness, sincerity" earned one point but an answer of "friendliness and sincerity" earned zero points). Prior to statistical analysis, a second researcher independently checked all data entry and scoring for accuracy. Scores were higher in the multiple-choice condition ($M = 4.52$, $SD = 0.80$) than the short-answer condition ($M = 2.37$, $SD = 1.20$), $t(121) = 11.55$, $p < .001$, $d = 2.14$.

After the test, participants completed a two-part survey. The first part of the survey consisted of three items modified from previous research to assess predictions about future performance on tests of knowledge (Fisher & Oppenheimer, 2021a). An item about free recall of information from the lesson asked "If you were to write down all the main ideas from the lesson from memory without any outside help or hints, what percentage of the information do you think you could correctly remember? (answer from 0% to 100%)." An analogous item about multiple-choice asked for an estimate of performance on a "multiple-choice test on this material in one week where you had to select the correct answers without any outside help or hints," and an item short-answer asked about an "open-ended test on this material in one week where you had to write in the correct answers without any outside help or hints."

The second part of the survey consisted of five standard teaching evaluation items adapted from the Student's Evaluation of Educational Quality (Marsh, 1983). The items included "the video lesson was clear," "the slides in the lesson were well prepared," "I learned and understood the lesson," "the quiz questions were an accurate and fair measure of what I learned from the lesson" and "overall, the lesson was excellent." Participants rated the items using a 5-point scale from *strongly disagree* to *strongly agree* with a midpoint of *neither disagree nor agree.* The items shared significant correlations ranging from .43 to .78 and their internal consistency was high ($\alpha = .89$). Moreover, the initial analyses indicated that trends were similar across the items. Specifically, no significant differences emerged between conditions after adjusting for multiple comparisons. As such, the Results section presents a total teaching quality score consisting of the mean of the five items.

Results

Predictions About Future Performance: Research Questions 1 and 2

The first question asked if students predict that they will perform better on multiple-choice tests, and the second asked how the test format affects predictions about future performance. The two research questions involved the same dependent variables. As such, to avoid redundancy and Type I errors, the analytic strategy was to answer both research questions using a single analysis. A 2×3 mixed analysis of variance (ANOVA) examined the between-subjects variable of testing condition (multiple-choice and short-answer) and the within-subjects variable of performance prediction (free-recall test, short-answer test, and multiple-choice test).[1] Means and standard deviations can be seen in Table 1. The effect of testing condition was significant, $F(1, 119) = 4.98$, $p = .027$, $\eta_p^2 = .040$, 95% CI [0.00, 0.13], and the within-subjects condition was significant, $F(2, 238) = 47.81$, $p < .001$, $\eta_p^2 = .287$, 95% CI [0.19, 0.37], but the interaction was not significant, $F(2, 238) = 2.90$, $p = .057$, $\eta_p^2 = .024$, 95% CI [0.00, 0.07]. Post hoc paired-samples t tests adjusted for multiple comparisons (.05/3) indicated that the within-subjects effect was significant because total mean performance prediction (see Table 1) was significantly higher for the multiple-choice test than for free recall or the short-answer tests by about 15%, all ts > 7.42, all ps < .001, all ds > 0.52, 95% CI [0.37, 0.67]. Because there was no interaction, the results indicate that perceptions of multiple-choice tests as relatively easy were present regardless of the type of test students had just taken. The main effect of the testing condition was significant because predictions about performance, overall, were 10% higher in the multiple-choice condition ($M = 72.48$, $SD = 21.82$) than in the short-answer condition ($M = 62.71$, $SD = 26.26$). In summary, participants

Table 1. Means and Standard Deviations for Predicted Future Performance in the Multiple-Choice and Short-Answer Test Conditions.

Predicted Performance	Total % *M* (*SD*)	Multiple-Choice Condition % *M* (*SD*)	Short-Answer Condition % *M* (*SD*)
Free recall	64.05 (27.45)	71.10 (24.01)	56.08 (29.08)
Short-answer test	61.54 (28.25)	65.40 (26.70)	57.34 (29.51)
Multiple-choice test	77.79 (25.26)	80.94 (22.32)	74.38 (27.91)

Note. Ratings occurred on a scale from 0% to 100%.

believed that a multiple-choice test would be easier than other tests, and taking a multiple-choice test led to increased predictions about later performance.

Perceptions of Teaching: Research Question 3

The third research question asked if test format affects perceptions of teaching effectiveness. To answer this question, an independent-samples t test compared ratings of teaching quality between the multiple-choice and short-answer test conditions. Evaluations were high in both conditions, with modal responses of 5 (*strongly agree*) across all items. The difference between the multiple-choice condition ($M = 4.47$, $SD = 0.80$) and the short-answer condition ($M = 4.26$, $SD = 0.75$) was not significant, $t(121) = 1.48$, $p = .141$, $d = 0.27$, 95% CI [−0.17, 0.64]. These results suggest that test format does not have a significant effect on students' evaluations of teaching.

Correlations With Test Performance: Research Questions 4 and 5

The fourth research question asked if test performance is related to predictions about future performance, and the fifth research question asked if test performance is related to perceptions of teaching effectiveness. To answer the research questions, Pearson's correlations examined the relations between test performance, predictions about future performance, and ratings of teaching quality. All correlations examined scores collapsed across conditions because trends were analogous in the multiple-choice and short-answer conditions. Results can be seen in Table 2. Test scores shared medium to large positive correlations with predictions about future performance. Test scores also shared medium-sized positive correlations with ratings of teaching quality. Thus, test performance was indeed related to predictions about future performance and perceptions of teaching effectiveness.

Discussion

The current research examined five research questions about the implications of using multiple-choice versus short-answer questions when testing students' memory for facts from a psychology lesson. The first research question asked if students predict that they will perform better on multiple-choice tests than on tests that require them to write down the answers. Participants did predict that their performance on a multiple-choice test would be about 15% higher than if they had to recall all information from the lesson freely or take a short-answer test. Students appear to understand that recognition-based multiple-choice questions are easier than recall-based short-answer questions. The second research question asked if the test format affects students' predictions about future performance. Test format did indeed affect predictions. Taking a multiple-choice test led to predictions about future performance that were 10% higher than taking a short-answer test. Compared to short-answer tests, the relative ease of multiple-choice tests appears to make students overconfident in their knowledge because they are less aware of the difficulty in recalling information from lessons without the hints provided by response options.

The third research question asked if test format affects perceptions of teaching effectiveness. The influence of test format on ratings of teaching quality was small and insignificant. The fourth research question asked if test performance is related to predictions about future performance. Students' scores on the test did share positive correlations with their predictions about how they would do on future tests where they were not allowed to use any form of outside assistance. From a metacognitive standpoint, students appear to generalize their performance on tests of immediate recall to future tests even if the future tests may be considerably more difficult. Finally, the fifth research question asked if test performance is related to perceptions of teaching effectiveness. Students who scored higher on the tests also tended to have more favorable attitudes about the quality of the lesson. Thus, even though test format did not significantly change ratings of teaching quality, higher test scores tend to predict higher student evaluations.

Table 2. Pearson's Correlations Between Test Score, Performance Predictions, and Evaluations of Teaching.

Variable	1	2	3	4	5
1. Test score	–				
2. Free-recall prediction	.60	–			
3. Short-answer prediction	.45	.70	–		
4. Multiple-choice prediction	.49	.70	.79	–	
5. Teaching quality	.37	.48	.46	.48	–

Note. All correlations are significant, $p < .001$.

Relation to Past Research

Results from the current study are consistent with past research on metacognition and student evaluations of teaching. Most prominently, this study extends the previously established effects of outside assistance on metacognition (Fisher & Oppenheimer, 2021a). In a previous study, hints during a test of memorized words led participants to inflate estimates of how many words they would remember on future tests. In the current study, hints in the form of response options on multiple-choice questions led to significantly increased estimates of how much they would remember on future tests about a psychology lesson where they received no hints or outside assistance. The current findings were also consistent with a large research base on student evaluations of teaching. Students tend to rate educational experiences positively if they perceive themselves as learning (Benton & Li, 2015; Stehle et al., 2012). This is consistent with the positive correlations that emerged in the current study between test scores, perceptions of learning, and evaluations of the lesson.

Strengths and Limitations

The implications of the current study must be considered in the context of its strengths and limitations. One major strength of the method was internal validity. Unlike classroom research where manipulations occur across different semesters or lessons, the only difference between conditions in the current study was the addition of response options for the multiple-choice tests. At the same time, the study also possessed external validity because the procedures closely mimicked how a lesson and comprehension check might occur in a real classroom. Conversely, the major limitation of the study was that it consisted of one short, prerecorded psychology lesson, and the results may not be generalized to full courses. In addition, the study occurred at a small, private, Midwestern university and needs to be replicated in different educational settings. Finally, this study focused on immediate recall and fact-based test questions. Results may be different for long-term retention or test questions that require higher-level cognitive skills from Bloom's taxonomy such as application or analysis (Krathwohl, 2002).

Implications

The current study suggests that teachers should consider students' metacognitive skills, not just learning, when taking advantage of the testing effect. Research on retrieval practice clearly indicates that both multiple-choice and short-answer tests can increase learning (Agarwal et al., 2021; McDermott et al., 2014). However, the current study and previous research suggest that providing outside assistance about the correct answer within test questions will affect students' predictions about future performance (Fisher & Oppenheimer, 2021a). Practically speaking, teachers who use fact-based multiple-choice test questions as a learning tool in the classroom could improve students' metacognition, and likely their memory, by converting the questions to open-ended format. In many cases, the conversion would be as easy as removing response options. Previous studies have demonstrated that the benefits of retrieval practice can occur even with ungraded assignments (Khanna, 2015; Lyle & Crawford, 2011). As such, the difficulty of providing graded feedback to individual students does not constrain the use of short-answer questions as a learning activity.

The current study also has implications for future research. Replication of the outside assistance effect is needed in real courses to determine if it generalizes across a range of instructors, lecture formats, test questions, and student evaluation surveys. Although the current study found no impact of test format on evaluations of teaching quality, future research should look for differences in the evaluation of teachers who rely on multiple-choice versus open-ended test questions on graded assignments, and it should also look for interactions between test format and factors known to affect teaching evaluations such as course topic, workload, and size (Benton & Cashin, 2012). Research is also needed to determine if these effects generalize to multiple-choice questions written at higher cognitive levels. Multiple-choice questions that require students to work with information in some way to obtain a correct answer may not affect metacognition the same way as questions that only require recognition of the correct answer. It will also be important for future research to examine the downstream effects of overestimating knowledge. Basic research on outside assistance indicates that the use of the internet to answer factual questions inflates confidence and leads people to store less information in memory, which then reduces their ability to answer later factual questions (Fisher et al., 2022). Analogously, students who initially test their knowledge using multiple-choice questions may prepare less for subsequent tests, leading to reduced performance.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Ethical Approval

All procedures received IRB approval. The materials, data, and syntax that support the findings of this study are openly available at https://osf.io/7txp2/.

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Open Practices

For publishing their material, Boysen and Osgood received badges for open data and open materials. The public content may be retrieved from https://osf.io/7txp2/.

Note

1. Descriptive analyses detected violations of normality, but exploratory nonparametic analyses showed the same trends as parametric analyses. We present analyses of means here for ease of interpretation.

References

Addison, W. E., Best, J., & Warrington, J. D. (2006). Students' perceptions of course difficulty and their ratings of the instructor. *College Student Journal*, *40*(2), 409–416.

Agarwal, P. K., Nunes, L. D., & Blunt, J. R. (2021). Retrieval practice consistently benefits student learning: A systematic review of applied research in schools and classrooms. *Educational Psychology Review*, *33*(4), 1409–1453. https://doi.org/10.1007/s10648-021-09595-9

Benton, S. L., & Cashin, W. E. (2012). *Student ratings of teaching: A summary of research and literature: IDEA paper #50*. The IDEA Center.

Benton, S. L., & Li, D. (2015). *Validity and reliability of IDEA "Teaching Essentials". IDEA Research Report #8*. IDEA Center.

Bjork, E. L., & Bjork, R. A. (2011). Making things hard on yourself, but in a good way: Creating desirable difficulties to enhance learning. In Gernsbacher, M. A., Pew, R. W., Hough, L. M., & Pomerantz, J. R. (Eds.), *Psychology and the real world: Essays illustrating fundamental contributions to society* (pp. 56–64). Worth Publishers.

Bjork, R. A., Dunlosky, J., & Kornell, N. (2013). Self-regulated learning: Beliefs, techniques, and illusions. *Annual Review of*

Psychology, *64*, 417–444. https://doi.org/10.1146/annurev-psych-113011-143823

Boysen, G. A., & Osgood, P. N. (2023). *Test format, learning overconfidence, and perceptions of teaching effectiveness* [Data set, syntax, materials]. https://osf.io/7txp2/.

Centra, J. A. (2003). Will teachers receive higher student evaluations by giving higher grades and less coursework? *Research in Higher Education*, *44*(5), 495–518. https://doi.org/10.1023/A:1025492407752

Cuddy, A. J. C., Fiske, S. T., & Glick, P. (2007). The BIAS map: Behaviors from intergroup affect and stereotypes. *Journal of Personality and Social Psychology*, *92*(4), 631–648. https://doi.org/10.1037/0022-3514.92.4.631

Dunlosky, J., & Rawson, K. A. (2015). Practice tests, spaced practice, and successive relearning: Tips for classroom use and for guiding students' learning. *Scholarship of Teaching and Learning in Psychology*, *1*(1), 72–78. https://doi.org/10.1037/stl0000024

Fisher, M., Goddu, M. K., & Keil, F. C. (2015). Searching for explanations: How the internet inflates estimates of internal knowledge. *Journal of Experimental Psychology: General*, *144*(3), 674–687. https://doi.org/10.1037/xge0000070

Fisher, M., & Oppenheimer, D. M. (2021a). Harder than you think: How outside assistance leads to overconfidence. *Psychological Science*, *32*(4), 598–610. https://doi.org/10.1177/0956797620975779

Fisher, M., & Oppenheimer, D. M. (2021b). Who knows what? Knowledge misattribution in the division of cognitive labor. *Journal of Experimental Psychology: Applied*, *27*(2), 292–306. https://doi.org/10.1037/xap0000310.supp

Fisher, M., Smiley, A. H., & Grillo, T. L. H. (2022). Information without knowledge: The effects of internet search on learning. *Memory*, *30*(4), 375–387. https://doi.org/10.1080/09658211.2021.1882501

Fiske, S. T., Cuddy, A. J. C., Glick, P., & Xu, J. (2002). A model of (often mixed) stereotype content: Competence and warmth respectively follow from perceived status and competition. *Journal of Personality and Social Psychology*, *82*(6), 878–902. https://doi.org/10.1037/0022-3514.82.6.878

Holzinger, A., Lettner, S., Steiner-Hofbauer, V., & Capan Melser, M. (2020). How to assess? Perceptions and preferences of undergraduate medical students concerning traditional assessment methods. *BMC Medical Education*, *20*(1), 1–7. https://doi.org/10.1186/s12909-020-02239-6

Jonides, J., Lewis, R. L., Nee, D. E., Lustig, C. A., Berman, M. G., & Moore, K. S. (2007). The mind and brain of short-term memory. *Annual Review of Psychology*, *59*, 193–224. https://doi.org/10.1146/ANNUREV.PSYCH.59.103006.093615

Khanna, M. M. (2015). Ungraded pop quizzes: Test-enhanced learning without all the anxiety. *Teaching of Psychology*, *42*(2), 174–178. https://doi.org/10.1177/0098628315573144

Krathwohl, D. R. (2002). A revision of Bloom's taxonomy: An overview. *Theory Into Practice*, *41*(4), 212–218. https://doi.org/10.1207/s15430421tip4104_2

Lyle, K. B., & Crawford, N. A. (2011). Retrieving essential material at the end of lectures improves performance on statistics exams. *Teaching of Psychology*, *38*(2), 94–97. https://doi.org/10.1177/0098628311401587

Marsh, H. W. (1983). Multidimensional ratings of teaching effectiveness by students from different academic settings and their relation to student/course/instructor characteristics. *Journal of Educational Psychology*, *75*(1), 150–166. https://doi.org/10.1037/0022-0663.75.1.150

McDermott, K. B., Agarwal, P. K., Antonio, L. D., Roediger, H. L., & Mcdaniel, M. A. (2014). Both multiple-choice and short-answer quizzes enhance later exam performance in middle and high school classes. *Journal of Experimental Psychology: Applied*, *20*(1), 3–21. https://doi.org/10.1037/xap0000004

Michaels, J. L. (2017). Quizzes benefit freshman and sophomore students more than junior and senior students in introductory psychology classes with noncumulative exams. *Scholarship of Teaching and Learning in Psychology*, *3*(4), 272–283. https://doi.org/10.1037/stl0000098

Narloch, R., Garbin, C. P., & Turnage, K. D. (2006). Benefits of prelecture quizzes. *Teaching of Psychology*, *33*(2), 109–112. https://doi.org/10.1207/s15328023top3302_6

Neto, J., Neto, F., & Furnham, A. (2022). Predictors of students' preferences for assessment methods. *Assessment and Evaluation in Higher Education*, *48*(4), 556–565. https://doi.org/10.1080/02602938.2022.2087860

Parkes, J., & Stefanou, C. (2010). Does pragmatism trump motivation in college students' preferences for examination formats? *Learning Environments Research*, *13*(3), 225–241. https://doi.org/10.1007/s10984-010-9077-4

Rivers, M. L. (2021). Metacognition about practice testing: A review of learners' beliefs, monitoring, and control of test-enhanced learning. *Educational Psychology Review*, *33*(3), 823–862. https://doi.org/10.1007/s10648-020-09578-2

Scouller, K. (1998). The influence of assessment method on students' learning approaches: Multiple choice question examination versus assignment essay. *Higher Education*, *35*(4), 453–472. https://doi.org/10.1023/A:1003196224280

Stehle, S., Spinath, B., & Kadmon, M. (2012). Measuring teaching effectiveness: Correspondence between students' evaluations of teaching and different measures of student learning. *Research in Higher Education*, *53*(8), 888–904. https://doi.org/10.1007/s11162-012-9260-9

Struyven, K., Dochy, F., & Janssens, S. (2005). Students' perceptions about evaluation and assessment in higher education: A review. *Assessment and Evaluation in Higher Education*, *30*(4), 325–341. https://doi.org/10.1080/02602930500099102